

Experiment 8: Tower of Hanoi

Aim

To implement the **Tower of Hanoi** problem using recursion in Python.

Algorithm

Step 1: Start.

Step 2: Read the number of disks n .

Step 3: If $n = 1$, move the disk from the source rod to the destination rod.

Step 4: Otherwise:

- Move $n-1$ disks from the source rod to the auxiliary rod.
- Move the largest disk from the source rod to the destination rod.
- Move the $n-1$ disks from the auxiliary rod to the destination rod.

Step 5: Display all the moves.

Step 6: Stop.

Source Code (Python)

```
main.py    Share  Run

1 # Tower of Hanoi
2
3 def hanoi(n, source, auxiliary, destination):
4     if n == 1:
5         print("Move disk 1 from", source, "to", destination)
6         return
7
8     hanoi(n - 1, source, destination, auxiliary)
9     print("Move disk", n, "from", source, "to", destination)
10    hanoi(n - 1, auxiliary, source, destination)
11
12 n = int(input("Enter number of disks: "))
13
14 print("Steps to solve Tower of Hanoi:")
15 hanoi(n, 'A', 'B', 'C')
```

Output

```
Output

Enter number of disks: 3
Steps to solve Tower of Hanoi:
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C

=== Code Execution Successful ===
```

Time Complexity

- Time Complexity: $O(2^n)$
 - Space Complexity: $O(n)$
-

Result

The **Tower of Hanoi** problem was implemented successfully in Python using recursion. The program displays the correct sequence of moves required to transfer all the disks from the source rod to the destination rod while following the rules of the puzzle.